Debugging and Exam Review

Lab 3

Can You Find the Bugs?

- The following program asks for the name and three exam scores, calculate the average, and display whether the student passed or failed.

- The program contains several errors.
  - Identify each error.
  - Explain why it is an error.
  - Correct the program.
  - Determine what the program should output.

Faulty Code:

```
#include <iostream>
#include <string>
using namespace std

double main()
{
    string name;
    int exam1, exam2, exam3;
    double average;

    cout >> "Enter your name: ";
    getline(cin, name);

    cout << "Enter Exam 1 score: ";
    cin << exam1;

    cout << "Enter Exam 2 score: ";
    cin >> exam2

    cout << "Enter Exam 3 score: ";
    cin >> exam3;

average = exam1 + exam2 + exam3 / 3;

    cout << "Student: " >> name << endl;
    cout << "Average: " << average << endl;

    if (average >= 70);
    {
        cout << "You passed!" << endl;
    }
    else
    {
        cout << "You did not pass." << endl;
    }

    return 0;
}
```

Can You Identify the Errors?

- Look carefully at the program.
  - Is every required semicolon ; present?
  - Is main() declared correctly?
  - Are << and >> being used correctly?
  - Is the arithmetic expression for average correct?
  - Is the if statement written correctly?
  - Are the variables using appropriate data types?
  - Are the braces correct?
  - Which errors are syntax errors?
  - Is there a logic error that might compile but produce the wrong answer?

Fix It Together

- using namespace std needs a semicolon using namespace std;
- double main() should be int main()
- cout >> "Enter your name: "; should be cout << "Enter your name: ";

The Logic Error

- average = exam1 + exam2 + exam3 / 3; Will this necessarily calculate the average correctly? No, because division has higher precedence.
  - Corrected statement average = (exam1 + exam2 + exam3) / 3.0;

Another Sneaky Bug

- if (average >= 70); What's wrong here?. The semicolon terminates the if statement.
- Corrected code is:

```
if (average >= 70)
{
    cout << "You passed!" << endl;
}
else
{
    cout << "You did not pass." << endl;
}
```

Correct Program

```
#include <iostream>
#include <string>
using namespace std;

int main()
{
    string name;
    int exam1, exam2, exam3;
    double average;

    cout << "Enter your name: ";
    getline(cin, name);

    cout << "Enter Exam 1 score: ";
    cin >> exam1;

    cout << "Enter Exam 2 score: ";
    cin >> exam2;

    cout << "Enter Exam 3 score: ";
    cin >> exam3;

    average = (exam1 + exam2 + exam3) / 3.0;

    cout << "Student: " << name << endl;
    cout << "Average: " << average << endl;

    if (average >= 70)
    {
        cout << "You passed!" << endl;
    }
    else
    {
        cout << "You did not pass." << endl;
    }

    return 0;
}
```

Find the Bugs: Student Profile

- Program Description
  - The program ask the user for their name, major, classification, age, and favorite programming language, then display a student profile.
  - The program contains several errors. Identify and correct them.

Faulty code:

```
#include <iostream>
#include <string>
using namespace std

int main()
{
    string name;
    string major;
    string classification;
    int age
    char favoriteLanguage;

    cout << "Enter your full name: ";
    getline(cin, name);

    cout << "Enter your major: ";
    getline(cin, major);

    cout << "Enter your classification: ";
    cin << classification;
    cout << "Enter your age: ";
    cin << age;

    cout << "Enter your favorite programming language: ";
    cin >> favoriteLanguage;

    cout << endl;
    cout << "----- Student Profile -----" << endl;
    cout << "Name: " >> name << endl;
    cout << "Major: " << major << endl;
    cout << "Classification: " << classification << endl;
    cout << "Age: " << age << endl;
    cout << "Favorite Programming Language: " << favoriteLanguage
    << endl
    return 0
}
```

Syntax errors

- `char favoriteLanguage;` should be `string favoriteLanguage;`
  - Because a `char` stores a single character.
- `cin >> favoriteLanguage;` should be `getline(cin, favoriteLanguage);`
- `cout << "Name: " >> name << endl;` should be `cout << "Name: " << name << endl;`
- `int age` should be `int age;`
- `cin << age;` and `cin << classification;` should be `cin << age;` and `cin >> classification;`
- `return 0` should be `return 0;`

Correct version

```
#include <iostream>
#include <string>
using namespace std;

int main()
{
    string name;
    string major;
    string classification;
    int age;
    string favoriteLanguage;

    cout << "Enter your full name: ";
    getline(cin, name);

    cout << "Enter your major: ";
    getline(cin, major);

    cout << "Enter your classification: ";
    getline(cin, classification);
    cout << "Enter your age: ";
    cin >> age;

    cin.ignore();

    cout << "Enter your favorite programming language: ";
    getline(cin, favoriteLanguage);

    cout << endl;
    cout << "----- Student Profile -----" << endl;
    cout << "Name: " << name << endl;
    cout << "Major: " << major << endl;
    cout << "Classification: " << classification << endl;
    cout << "Age: " << age << endl;
    cout << "Favorite Programming Language: "
    << favoriteLanguage << endl;

    return 0;
}
```

Practice: Can you identify and fix all the errors?

```
#include <iostream>
#include <iomanip>
using namespace std

int main()
{
    double length, width;
    double area, perimeter

    cout << "Enter the length: ";
    cin << length;

    cout << "Enter the width: ";
    cin >> width;

    area = length + width;
    perimeter = 2 * length + width;

    cout << fixed << setprecision(2);

    cout >> "Area: " << area << endl;
        cout << "Perimeter: " << perimeter << endl

        return 0;
}
```

Solution

```
#include <iostream>
#include <iomanip>
using namespace std;

int main()
{
    double length, width;
    double area, perimeter;

    cout << "Enter the length: ";
    cin >> length;

    cout << "Enter the width: ";
    cin >> width;

    area = length * width;
    perimeter = 2 * (length + width);

    cout << fixed << setprecision(2);
    cout << "Area: " << area << endl;
    cout << "Perimeter: " << perimeter << endl;

    return 0;
}
```